

SAGE Difference Report

Where we were (IRL SAGE PAPER)?

- **Core Architecture**
 - Dual-Store Database
 - Stateless Sessions
- **The 3-Layer Safety Gatekeeper**
 - Role-Based Access Control (RBAC)
 - Attribute-Based Access Control (ABAC)
 - Cognitive Access Control (CAC)
- **Local NLP models**
 - Intent Classifier
 - Entity Extraction
- **Teacher Controls**
 - Topic Toggles

SRL SAGE (Software Feature)

- **Data & Memory Architecture**

- **Tri-Database Architecture:** Added SQLite to act as the persistent "Memory" and "State" of the student, linking with ChromaDB and Neo4j.
- **Bayesian Knowledge Tracing (BKT):** Replaced binary (locked/unlocked) flags with deep probability math. Tracks mastery exactly across three tiers: Quiz (Declarative), Micro-coding (Procedural), and Code Review (Applied).
- **Forgetting Curve (SM-2):** Mathematically degrades a student's mastery score over time if they don't practice.
- **Misconception Tracking:** Uses an Exponential Moving Average (EMA) to flag when a student is highly confident but factually wrong.

SRL SAGE (Software Feature)

- **Cognitive & Affective Modeling (Emotion)**
 - **Affective Computing (Emotion Tracking):** Uses a local model to read student frustration in real-time, calculating a "Frustration Delta" (ΔF).
 - **Cognitive Load Tracking:** Analyzes the complexity of the student's pasted code (AST depth proxy).
 - **Attention Hijacking Tracker:** Tracks "off-topic strikes" if a student tries to distract the AI.

SRL SAGE (Software Feature)

- **Multi-Agent Orchestration**

- **Sentinel Agent (Security):** Uses math formulas (S_{cog} , S_{goal} , S_{acad} , S_{spam}) to instantly block queries if the student is emotionally overwhelmed, cheating, or off-topic.
- **Scaffolding Agent:** Intercepts "build a program" requests. Refuses to write code. Instead, generates an asynchronous 3-5 step JSON plan and forces the student to write the code step-by-step.
- **Examiner Agent:** Triggers proactive "Brain Warm-Up" pop-quizzes based on memory decay. Uses Dual-Layer Grading (Fast local vectors + LLM judge fallback).
- **Reviewer Agent:** Audits pasted code using the "Sandwich Method". Includes a "**Rigorous Analysis**" mode that acts as a Senior Engineer looking for memory leaks and edge cases.
- **Socratic Agent:** The standard fallback. Uses Graph-expanded RAG and generates dynamic Mermaid.js flowcharts.

SRL SAGE (Software Feature)

- **Human-in-the-Loop & Self-Calibration (Core Novelty)**
 - **SRL Dashboard Sliders:** Students can manually drag a slider to lower their AI-calculated mastery score if they feel they were just "guessing."
 - **Auto-Tuning Algorithms:** If the system detects a student manually lowering their score frequently, it automatically rewrites its own Bayesian parameters (P_G / Guess Rate) to become a stricter grader for that user.
- **Personalization Engine**
 - **Teaching Style Bandit (UCB1):** A reinforcement learning algorithm that tests different teaching styles (Analogy vs. Technical vs. Visual) and optimizes for the style the student "Thumbs Ups" the most.
 - **Mastery-Conditioned Prompting:** The LLM prompt changes drastically based on BKT scores (e.g., heavy hand-holding for "Novices" vs. concise refreshers for "Reviewing" students).

SRL SAGE (Software Feature)

- **New User Modalities & Interfaces**

- **Context-Aware Video Classroom:** Teachers upload videos. Local Whisper AI transcribes them. When a student pauses to ask a question, the LLM is *only* fed the transcript up to that exact timestamp to prevent spoilers.
- **Gamified Student Dashboard:** Generates dynamic "Skill Trees" (Learning Paths) based on a student's personal project goal (e.g., "Build a Calculator"). Includes XP tracking and Radar Charts.
- **Micro-Challenge Queue:** If a student skips an AI coding challenge, it goes into a "Pending Tasks" dropdown so they can't escape the work.
- **Accessibility & Neurodiversity Suite:** Toggles for Dyslexia-friendly fonts, high-contrast code, extra line spacing, and 45-minute ADHD break reminders.
- **Teacher Command Center:** A master dashboard calculating student "Risk Levels" (combining inactivity, frustration, and low BKT scores).

SRL SAGE (Software Feature)

- **Production Engineering & Resilience**

- **Mermaid Sanitizer:** Regex engine that stops LLM hallucinations from crashing the UI flowchart renderer.
- **Garbage Concept Blocklist:** A self-learning cache that stops the database from indexing meaningless words (like "ok" or "thanks").
- **Goal Poisoning & Path Traversal Guards:** API-level security preventing students from hacking the semantic radius and teachers from uploading malicious file paths.
- **Adaptive SSE Buffer:** A UI mechanism that speeds up or slows down the AI's "typing" animation based on network jitter.

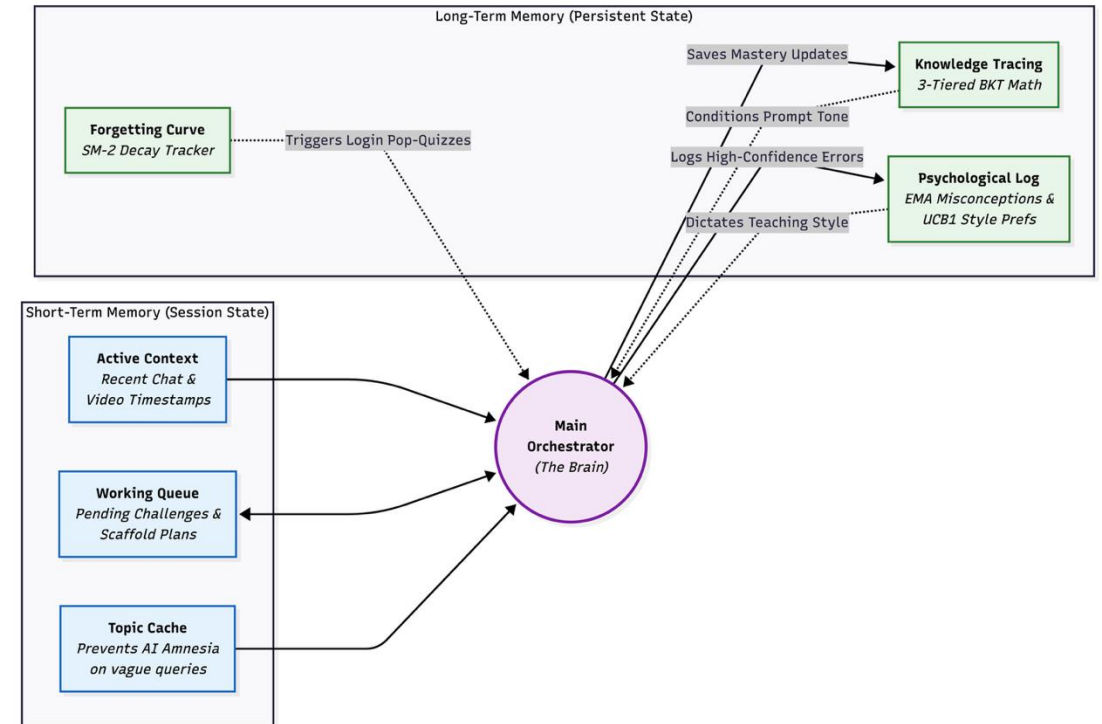
What these software features enable?

- **User Memory/context (Short-Term & Long-Term)**
- **Multi-Dimensional Learner Model**
- **Self-Regulated Learning**
- **Hyper-Adaptive Personalization**
- **Autonomous System Calibration**
- **Cognitive & Academic Security**

User Memory/context

The Dual-Memory Architecture enables *Pedagogical Continuity and Adaptive Pacing*

- Key advantages:
 - **Prevents "Context Escape"**
 - tracks pending tasks, so students can't skip difficult questions
 - **Prevents "AI Amnesia"**
 - Topic cache remembers the current discussion
 - **Prevents "Illusion of Competence"**
 - SM-2 reinforces learning over time
 - **Enables Hyper-Personalization (LTM)**
 - Long-term memory stores misconceptions and learning preferences to personalize teaching



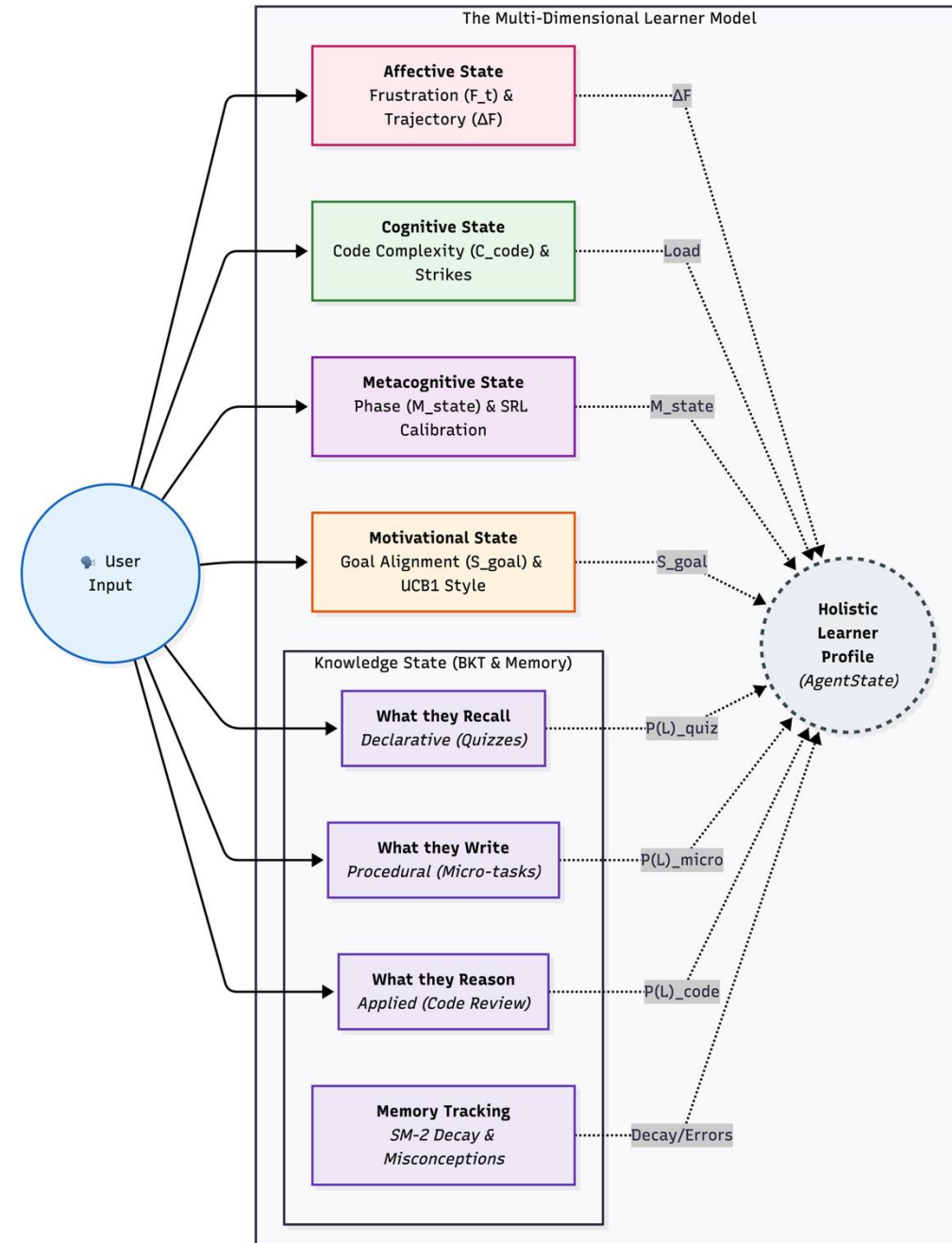
Learner Model

Shows how the AI understands not just *what* a student asks, but *how* they learn, think, and struggle in real time.

- **5 Real-Time Psychological Pillars:**

- *Affective*: Emotion & Frustration (ΔF)
- *Cognitive*: Workload & Attention
- *Metacognitive*: Self-Awareness & Phase
- *Motivational*: Long-Term Goals
- *Knowledge*: 3-Tier BKT & Memory Decay

- **The Result:** Two students asking the exact same question will get completely different AI responses.



Deep Integration of Self-Regulated Learning (SRL)

Enforcing the "How to Learn" Cycle

- **Forethought (Planning)**
 - **Goal-Driven Skill Trees:** Breaks learning goals into prerequisite skills.
 - **Pre-Code Logic Prompts:** Students explain their solution before coding.
- **Performance (Monitoring)**
 - **Pending Task Queue:** Tracks skipped challenges to prevent avoidance.
 - **Real-Time Proficiency Radar:** Visualizes strengths and weak skill areas.
- **Self-Reflection (Evaluation)**
 - **Pre-Quiz Confidence Ratings:** Compares predicted confidence with actual performance.
 - **Metacognitive Overrides:** Students can adjust their own mastery estimates.
 - **Algorithmic Auto-Tuning:** Learner feedback refines future mastery predictions.

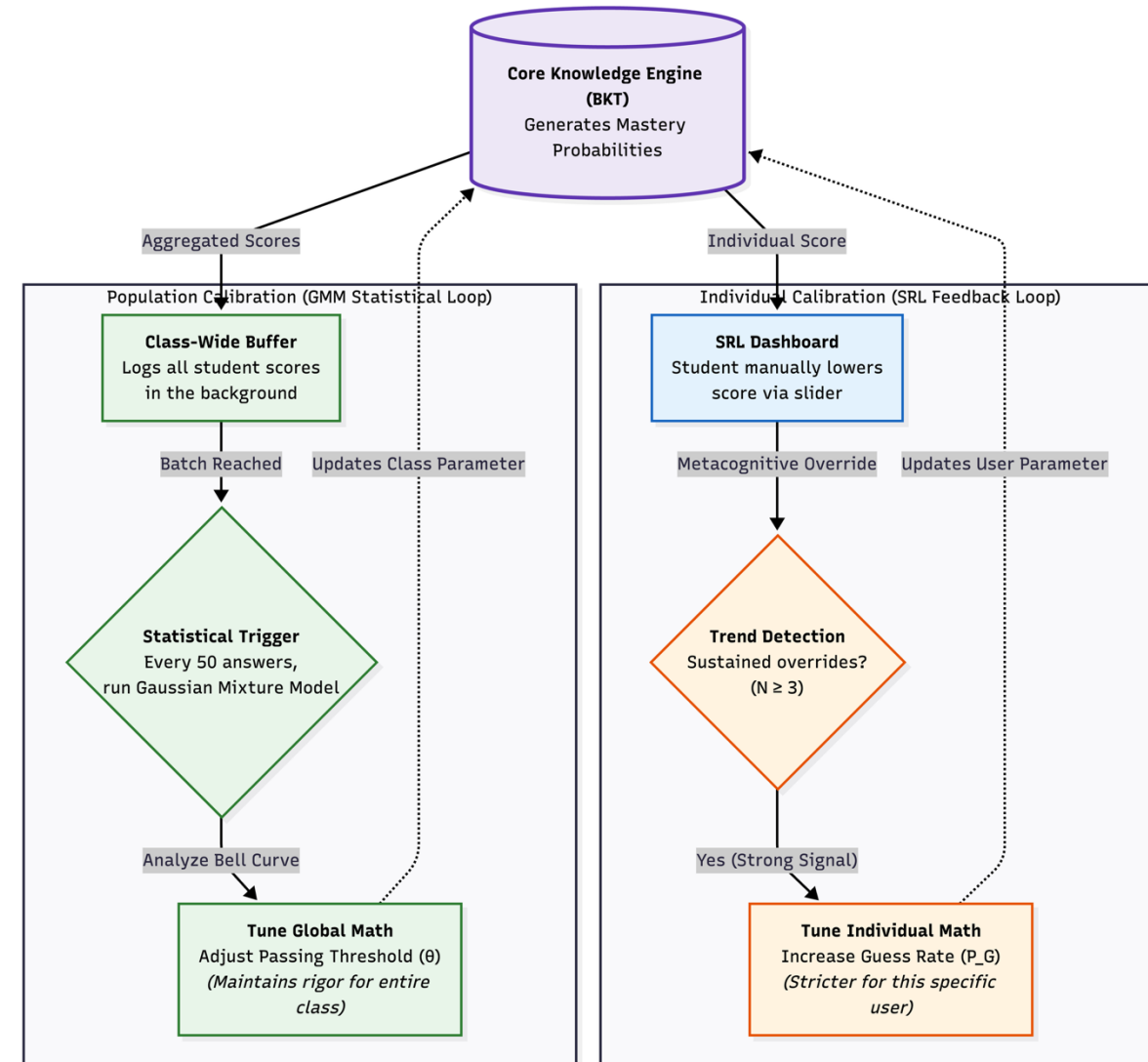
Hyper-Adaptive Personalization (Dynamic Scaffolding)

- **Enabled by:** The **UCB1 Teaching Style Bandit** (finding out if a student learns better via analogies or visual diagrams) and **Mastery-Conditioned Prompting**.
- **The Result:** A "Novice" who is highly frustrated (ΔF spike) gets a completely different AI persona (empathetic, no code, heavy hand-holding) compared to a "Proficient" student who is calm (concise, technical, edge-case focused).

User Memory/context

Tutor continuously updates its grading parameters through two self-correcting feedback loops.

- **Enabled By**
 - **Individual Loop:** SRL Auto-Tuning (student feedback)
 - **Population Loop:** GMM Threshold Calibration (class-wide statistics)
- **The Result**
 - **Individual:** Students who frequently admit they were guessing receive stricter future mastery estimates.
 - **Population:** The system adjusts the global passing threshold based on overall class performance to maintain consistent difficulty.



Cognitive & Academic Security (Safe AI Teaming)

- *This goes far beyond "**Prompt Injection**" defense. It protects the student's brain and the teacher's curriculum.*
- **Enabled by: The Sentinel Agent's and the Reviewer Agent's Sandwich Method.**
- **The Result:**
 - The AI refuses to dump code (Academic Integrity).
 - It locks the chat if the student is severely overwhelmed (Cognitive Security).
 - It issues strikes if the student tries to talk about off-topic subjects (Attention Security).

Key Novelty

- **Factored Multi-Tier (Bayesian Knowledge Tracing) BKT with Conjunctive Certification** ★
 - Multi-tier evidence routing across Bloom's taxonomy.
 - Online BKT updates with mastery ceilings.
 - Conjunctive certification ensures diverse evidence before mastery.
- **Mastery-Conditioned Response Adaptation**
 - AI adapts explanations, hints, and difficulty based on mastery.
 - **Established ITS practice** that demonstrates how the mastery model is applied.
- **SRL-BKT Calibration Loop** ★
 - Students can challenge their mastery estimates.
 - Sustained self-corrections update personal BKT parameters (guess/slip rates).
 - Produces a personalized, increasingly accurate learner model over time.

What needed to be done Fall Class trial?

For SRL Data collection

The SRL Data We Can Collect

- **Measuring Forethought (Planning):**
 - We track when students set **Macro-Goals** and measure adherence to AI-generated Skill Trees.
- **Measuring Performance & Self-Control (Monitoring):**
 - We log **Cognitive Debt** (skipped vs. completed challenges).
 - We track "**Lazy Learner**" behavior (copying code vs. manual typing).
 - We classify every chat turn into an **SRL state** (Planning, Monitoring, Reflecting, or Helplessness).
- **Measuring Self-Reflection & Metacognition (Evaluation):**
 - **Judgment of Learning (JOL):** Students rate confidence before quizzes, enabling confidence vs. performance analysis.
 - **Calibration Metric:** We log when students lower their AI-estimated mastery, enabling personalized BKT calibration.
 - We capture **prediction-before-outcome** data to evaluate calibration accuracy over time.
- **Measuring Cognitive & Emotional Regulation:**
 - We track **Frustration Trajectories (ΔF)** and relate them to behaviors such as skipping tasks or excessive copying.
 - We monitor **Misconception Trajectories**, measuring how misconceptions form and are resolved.
 - We capture **item-level quiz psychometrics**, including responses, confidence, and time-to-answer.

Baseline & Onboarding (Week 1)

- **Goal:** Establish the ground truth and initiate the SRL *Forethought* phase.
- **Instructor Action 1: Pre-Assessments**
 - Administer the C-Programming Pre-Test and the Pre-MAI (Metacognitive Awareness Inventory) survey.
 - *Data Captured:* assessment (Baseline Bloom-tagged knowledge and SRL traits).
- **Instructor Action 2: Macro-Goal Setting**
 - Assign students to log into SAGE and set a specific project goal (e.g., "Build a text adventure game").
 - *Data Captured:* user_goals and event_log(goal_set). Initiates the *Forethought* phase.
- **Instructor Action 3: Topic Lock**
 - Lock all topics except the Week 1 fundamentals (e.g., "Variables").
 - *Data Captured:* Baseline path_event logging begins.

Declarative Knowledge & JOL Baseline (Weeks 2–3)

- **Topics:** Variables, Data Types, Basic I/O
Goal: Train students to use the AI and establish a baseline for their Metacognitive accuracy (Judgment of Learning).
- **Instructor Action 1: Unlock New Topics Sequentially**
 - As Instructor teaches Control Flow/Loops, unlock them in the Dashboard.
- **Instructor Action 2: Mandate Pop-Quizzes (Low-Stakes)**
 - Tell students to always complete the AI-prompted quizzes.
 - *Data Captured: Lens 3 (SRL Evaluation).* We capture massive amounts of jol_log (Confidence 1-5 vs. Actual Correctness) and quiz_log (Time-to-answer). We establish if students are initially overconfident.
- **Instructor Action 3: Natural Chat Interaction**
 - Allow students to ask standard questions.
 - *Data Captured: Lens 1 (Learner Model).* turn_log begins tracking the m_state FSM (Planning/Monitoring) and baseline affect_log (frustration). bkt_history begins populating the **Quiz Tier** posteriors.

Productive Struggle & Calibration (Weeks 4–5)

- **Topics:** Control Flow, Loops, Logic
Goal: Induce cognitive friction. This is where topics get harder, frustration spikes, and we introduce the *Calibration Loop (Core Novelty 2)*.
- **Instructor Action 1: The "5-Minute Reflection" (Crucial for the Paper)**
 - Once a week, Instructor tells the class: *"Look at your SAGE Dashboard. If the AI says you have 90% mastery in Loops, but you feel lost, pull the slider down."*
 - *Data Captured: Lens 2 (BKT Calibration).* This triggers `calibration_log` (slider moves, Δ , direction EMA). This is the exact moment we log the `prediction_log` (predicting their next quiz score *before* the outcome based on adapted P_G).
- **Instructor Action 2: Monitor "Cognitive Debt"**
 - Assign a coding task. SAGE will trigger micro-challenges. Instructor reminds students to clear their "Pending Tasks" queue.
 - *Data Captured:* `behavior_log` (Thumbs down, copy_code clicks) and `event_log(quiz_skip)`. We track *Performance/Monitoring* SRL behaviors.
- **Instructor Action 3: Affective Tracking**
 - As loops get difficult, students will get frustrated.
 - *Data Captured:* `affect_log` (ΔF spikes) and `misconception_log`. We track how high-confidence errors form and decay via EMA.

Application, Review, & Memory Decay (Weeks 6–7)

- **Topics:** Functions, Arrays, Pointers
Goal: Test the "Evidence Diversity Guarantee" (Code Tier BKT) and the SM-2 Forgetting Curve.
- **Instructor Action 1: Heavy Code Submissions**
 - Assign homework requiring students to paste raw C code into SAGE for review.
 - *Data Captured:* evidence_log moves from the Quiz tier to the **Micro and Code tiers**. We track if students who passed quizzes in Phase 1 can actually write the code now, proving the *Conjunctive Certification* novelty (Core Novelty 1).
- **Instructor Action 2: Let Memory Decay Naturally**
 - Instructor does nothing here. Because 4 weeks have passed since "Variables," SAGE's SM-2 algorithm will automatically force "Brain Warm-Up" quizzes on old topics.
 - *Data Captured:* session_log and bkt_history. We track the trajectory of decayed $P(L)$ bouncing back upon successful review.
- **Instructor Action 3: Push Custom Challenges**
 - Instructor uses the Teacher Dashboard to push challenges to students showing high delta_f (frustration) and low mastery.
 - *Data Captured:* response_log and assignments. Proves how teacher-in-the-loop intervention stabilizes the m_state (moving students from *Helplessness* back to *Planning*).

Culmination & Data Synthesis (Week 8+)

- **Topics:** Final Project / Final Exam Prep
Goal: Close the loop and capture the final ground truth.
- **Instructor Action 1: Unlock All Topics**
 - Students have unrestricted access to build their final projects.
 - *Data Captured:* path_event (Did they follow their original goal set in Phase 0?).
- **Instructor Action 2: Post-Assessments**
 - Administer the Post-Test (C-Programming) and Post-MAI Survey.
 - *Data Captured:* assessment.

Expected Academic Contributions

Factored Multi-Tier BKT with Conjunctive Certification

- **The Claim:** Tracking mastery across three independent cognitive tiers (Declarative, Procedural, Applied) and requiring *Conjunctive Certification* (passing all three) is a vastly superior predictor of true learning than a standard, flat BKT model.
- **How the Data Proves It:**
 - **The Classroom Action:** Instructor assigns SAGE as the primary tool (Action 1) and pushes targeted micro-challenges (Action 5), forcing students to generate data across all three tiers: Quizzes (Declarative), Micro-coding (Procedural), and raw Code Reviews (Applied).
 - **The Ground Truth Action:** Instructor administers a Pre- and Post-Test (Action 6) specifically structured with questions matching those three Bloom's tiers.
 - **The Analysis:** We will extract the final $P(L)$ (Probability of Learning) scores from the SQLite `user_knowledge` table for all three tiers individually, as well as the combined *Conjunctive Certification* flag. We will run a correlation analysis against their actual Post-Test grades.
 - **The Proof:** We expect the data to show that students who achieved a high score in *only one tier* (e.g., Quiz) but failed the others performed poorly on the exam. Conversely, students who triggered the *Conjunctive Certification* (proving the "Evidence Diversity Guarantee") will show the highest Normalized Learning Gains, proving that multi-tier BKT is necessary to prevent the "illusion of competence."

The SRL-BKT Calibration Loop

- **The Claim:** Allowing students to express sustained metacognitive disagreement (via a downward-only slider) to actively tune the AI's Bayesian Guess Rate (P_G) creates a more accurate, individualized learner model than behavioral data alone.
- **How the Data Proves It:**
 - **The Classroom Action:** Instructor implements the weekly 5-Minute "Dashboard Reflection Days" (Action 2), explicitly prompting students to evaluate their radar charts and physically use the downward-only SRL sliders if the AI is overestimating them.
 - **The Analysis:** We will query the user_bkt_calibration table to isolate two groups:
 - *Group A (Active Calibrators):* Students who registered ≥ 3 downward adjustments, successfully triggering the algorithmic auto-tuning of their P_G parameter.
 - *Group B (Passive Users):* Students who never adjusted their sliders, leaving the P_G at the population default.
 - **The Proof:** We will compare Group A and Group B using the Metacognitive Awareness Inventory (MAI) survey results (Action 6) and their Post-Test scores. We expect the data to prove that Group A (who trained the algorithm to be a stricter grader) achieved significantly higher long-term retention and demonstrated measurable growth in metacognitive awareness compared to the passive group.

Mastery-Conditioned Response Adaptation

- **The Claim:** While adapting AI prompts based on student mastery is standard ITS behavior, doing so using our highly accurate, SRL-calibrated, multi-tier BKT signal drastically improves the effectiveness of the AI's scaffolding, enforcing productive struggle.
- **How the Data Proves It:**
 - **The Classroom Action:** Instructor locks the curriculum week-by-week (Action 4) and uses Pop-Quizzes for low-stakes grades (Action 3). This forces students to rely heavily on SAGE's hints and scaffolding when they encounter locked or struggling concepts, generating massive logs in the messages table.
 - **The Analysis:** Because the LLM's prompt physically changes based on the student's BKT level (e.g., "Novice" gets heavy scaffolding, "Proficient" gets edge cases), we will track the behavioral outcomes in the chat logs *immediately following* the AI's response.
 - **The Proof:** We will measure "Productive Struggle" metrics:
 - Does the frequency of "Copy Code" clicks drop when a student transitions from Novice to Developing?
 - Does the Dwell Time (response latency) increase when the AI uses Proficient-level prompts?
 - Do off-topic strikes and bypass attempts decrease over the 30 days?
 - By showing that the students' behavior shifts positively as their BKT-driven classification evolves, we prove that our underlying BKT and SRL signals are highly accurate triggers for standard adaptive scaffolding.